

SGTX Platform

Master Blueprint

Clean Master Edition v14.0

Sovereign Governed Trade Execution Infrastructure

STATUS: AUDITED / INTEGRATED / CANONICAL

Document Date: 2026-08-26

Prepared by: Master Blueprint Integration Engine

Classification: Internal Technical Master Specification

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Executive Summary — What Changed from v13.1 and Why

This Clean Master Edition v14.0 restructures the SGTx Platform Master Blueprint into three strict layers: Layer 0 (Constitution), Layer 1 (Architecture), and Layer 2 (Implementation Specifications). The v13.1 baseline is treated as the audited historical record; all changes in v14.0 are additive, clarifying, or restructuring. No material, principle, or capability has been silently deleted.

The primary motivation for v14.0 is to produce a document that a bank, CBE-class regulator, institutional investor, or independent engineering team can treat as a single source of truth. The v13.1 document, while comprehensive, mixed immutable constitutional principles with implementation details, used inconsistent deployment-state vocabulary, and contained residual over-claim language ('production-ready', 'complete', 'zero-cost') that is inappropriate for a regulated multi-jurisdiction platform.

Key changes in v14.0: (1) Three-layer separation with every normative statement tagged [L0], [L1], or [L2]. (2) Elimination of all over-claim language — replaced with precise deployment-state vocabulary: CORE_READY, PRODUCTION_CONNECTED, LEGAL_AUTHORIZATION_REQUIRED. (3) The 24 audit findings from v13.1 Part A (A-01 through A-24) are enforced as governing language, not re-opened. (4) Full 28-add-on catalogue with explicit status matrix. (5) 4-dimension external readiness reporting (TECHNICAL / LEGAL / OPERATIONAL / COMMERCIAL). (6) Command/Event taxonomy formalized. (7) Historical audit trail preserved as immutable annex.

What v14.0 is NOT: It is not a new architecture. It is not a redesign. It does not introduce new capabilities beyond what v13.1 specified. It is a clean, restructured, institutionally-defensible presentation of the same architecture, with all contradictions resolved, all over-claims eliminated, and all principles clearly separated by mutability layer.

LAYER 0 — CONSTITUTION

[L0] This layer contains immutable principles only. These principles are the non-negotiable foundation of the SGTX platform. They may be amended only via an explicit multisig + notice process (see §0.4 Amendment Process). No implementation decision, business pressure, or operational convenience may override a Layer 0 principle.

0.1 Governor Principles (G1–G7)

[L0] The Governor is the constitutional enforcement engine of SGTX. Seven principles govern its operation:

Principle	Statement
G1 — Execution Always Gated	Every irreversible action requires Governor approval before execution. No component may bypass the Governor pipeline.
G2 — OPA Enforced	Open Policy Agent (OPA) evaluates every decision against authored policies. Policy violations block execution.
G3 — WasmEdge Constitutional	WasmEdge executes constitutional rules (non-custody, non-marketplace, 110% reserve, closure-is-earned) as deterministic WebAssembly modules. These cannot be overridden by configuration.
G4 — Loom Audited	Every Governor decision is appended to the Loom (SHA-256 hash-chained audit log). The Loom is immutable and externally verifiable.
G5 — Multisig for Irreversible	Irreversible actions (USTN closure, policy amendment, fund release) require multisig approval (2-of-3 standard, 3-of-5 constitutional).
G6 — AI Advisory Only	The AI subsystem (A1–A3) may propose, explain, classify, and escalate, but NEVER has independent execution authority. A4 automation is deterministic policy execution, not AI autonomy.
G7 — Bank-Authoritative Settlement	SGTX orchestrates settlement instructions but never becomes the settlement authority. Banks and regulated financial institutions confirm settlement finality. SGTX is non-custodial by architecture.

0.2 The 32-Point SGTX Transaction Constitution

[L0] The following 32 points constitute the immutable constitution of the SGTX platform. Every trade, every component, and every integration must comply with all 32 points.

[L0] 1. Non-custodial — SGTX never holds customer funds. FeeLock escrow is non-custodial; funds remain with the regulated bank/PSP until settlement is confirmed.

[L0] 2. Non-marketplace — SGTX does not match buyers with sellers. Trades occur between known, relationship-controlled counterparties.

[L0] 3. Non-title-taking — SGTX never takes title to goods. Title transfer is governed by the contract Incoterm and applicable law.

[L0] 4. Non-carrier — SGTX is not a carrier. It orchestrates logistics execution through approved carriers, LSPs, and shipping lines.

[L0] 5. Non-customs-authority — SGTX is not a customs authority. It interfaces with customs authorities (Nafeza, single-window systems) but never replaces them.

[L0] 6. Non-bank — SGTX is not a bank. It orchestrates payment instructions but never holds deposits or executes settlement.

[L0] 7. Non-deposit-taking — SGTX does not accept deposits. All funds flow through connected banks and regulated PSPs.

[L0] 8. Non-government — SGTX is not a government entity. It serves as infrastructure connecting commercial parties with government systems.

[L0] 9. AI-assisted — AI provides advisory (A1), constraining (A2), and escalation (A3) capabilities. A4 automation is deterministic policy execution.

[L0] 10. Governor-governed — Every irreversible action passes through the Governor pipeline (G1–G7).

[L0] 11. OPA-enforced — Open Policy Agent evaluates every decision against authored policies.

[L0] 12. WasmEdge-enforced — Constitutional rules execute as deterministic WebAssembly modules.

[L0] 13. Loom-audited — Every decision is appended to the SHA-256 hash-chained Loom audit log.

[L0] 14. USTN-centric — The Universal Shipment Tracking Number (USTN) is the canonical namespace for every trade. All downstream records reference the USTN.

[L0] 15. Jurisdiction-aware — Every trade is evaluated against the applicable jurisdiction's regulatory profile, customs procedure, and legal framework.

[L0] 16. Relationship-controlled — Counterparty relationships are explicitly established (approved, connected, saved GTID). No random matching, no public rankings.

[L0] 17. 110% reserve rule — If reserve metadata is maintained, it must be at least 110% backed. The constitutional layer sets the threshold; ZK attestation provides evidence.

[L0] 18. Closure-is-earned — A USTN is closed only when all 7 closure conditions are met (delivery accepted, settlement complete, reconciliation complete, customs complete, post-clearance complete, disputes satisfied, evidence sealed). Closure cannot be forced.

[L0] 19. Recovery ≠ erasure — Recovery actions (exception resolution, obligation failure cascade) restore state but never erase audit history. The Loom is immutable.

[L0] 20. USTN as namespace, not override — The USTN is a universal reference namespace. It does not override external authoritative systems (government references, bank transaction IDs, customs declarations).

[L0] 21. Bank-authoritative settlement — Banks and regulated financial institutions are authoritative for money movement and settlement confirmation. SGTX orchestrates instructions; it does not confirm settlement.

[L0] 22. Non-custody is architectural — Non-custody is an architectural property of SGTX, not a standalone legal classification. Actual functionality determines licensing in each jurisdiction.

[L0] 23. Reserve metadata ≠ custody — Reserve tables store attestations, ratios, and evidence metadata. They do not create customer-fund custody.

[L0] 24. Stablecoin/DeFi conditional — Stablecoin/DeFi rails are conditional, jurisdiction-permitted financing capabilities, not canonical settlement authority. They are sub-rails beneath bank-authoritative settlement.

[L0] 25. GNN non-marketplace bounded — The Graph Neural Network (Add-On 1) provides trust analytics for known parties. It never discovers, recommends, ranks, or introduces counterparties.

[L0] 26. Direct API = first-party connector — 'Direct API' denotes a currently adopted first-party connector (e.g., Nafeza, CargoX, ETA, CBE). The worldwide adapter fabric is the extensibility layer.

[L0] 27. RoRo is first-class — Roll-on/Roll-off cargo is a first-class transport mode, not a sub-mode of ocean container. It has its own entity types, state machines, and terminal adapters.

[L0] 28. Mode-specific government applicability — Government integrations (e.g., Egypt Nafeza) have mode-specific applicability. A single generic maritime workflow is insufficient; ROAD, AIR, OCEAN_CONTAINER, RORO, RAIL each have distinct customs procedures.

[L0] 29. External readiness is 4-dimensional — Every integration reports independently: TECHNICAL, LEGAL, OPERATIONAL, COMMERCIAL. 'Connected' requires all four dimensions.

[L0] 30. Production-readiness vocabulary — Use CORE_READY, ADAPTER_READY, COUNTRY_CONFIGURED, SANDBOX_CONNECTED, PRODUCTION_CONNECTED, LEGAL_AUTHORIZATION_REQUIRED, MANUAL_ONLY, PORTAL_ONLY, INTEGRATION_REQUIRED. Never claim WORLDWIDE_INTEGRATED without evidence.

[L0] 31. Manual fallback is governed — Manual fallback (API → EDI → SFTP → PORTAL → BROKER → MANUAL) is authenticated, attributable, timestamped, hashed, and Loom-logged.

[L0] 32. Evidence is sealed at closure — The final evidence package (26 categories per Art 101) is sealed at USTN closure. Post-closure observation (§22) may add evidence but cannot modify sealed evidence.

0.3 AI Authority Ladder (A0–A5)

[L0] The AI subsystem operates on a strict authority ladder. A5 is FORBIDDEN.

Level	Name	Authority	Boundary
A0	None	No AI involvement	Pure deterministic rules
A1	Advisory	May explain, translate, suggest, summarise, notify, generate draft instructions	Never makes decisions; never blocks; never forces
A2	Constraining	May classify, detect anomalies/discrepancies, predict delays, compare images, estimate ETA, optimise ULD/stowage, analyse route	Proposes constraints; Governor decides whether to enforce
A3	Escalation	May escalate to human review, flag for enhanced due diligence, trigger compliance review	Escalates; never resolves autonomously
A4	Execution (within bounds)	Deterministic policy automation executed by Governor + OPA + WasmEdge under pre-authorised rules	AI NEVER acquires independent execution authority; A4 is policy, not AI autonomy
A5	FORBIDDEN	Autonomous AI decision-making without human authorization	CONSTITUTIONALLY PROHIBITED. No component may implement A5.

0.4 Amendment Process

[L0] Layer 0 principles may be amended only through the following process: (1) A formal amendment proposal is submitted to the Platform Governance Authority. (2) The proposal undergoes a mandatory 30-day notice period during which all affected parties may review and comment. (3) The amendment requires 3-of-5 multisig approval from the constitutional signatories. (4) The amendment is appended to the Loom with its full provenance (proposer, reviewers, signatories, timestamp). (5) The previous version is NEVER deleted; it is marked as superseded with a forward reference to the new version. This process ensures full traceability — no constitutional change is ever silent.

LAYER 1 — ARCHITECTURE

[L1] This layer contains the current normative architecture of SGTx. These specifications describe how the platform is structured and how components interact. They are mutable through the standard change-management process (see Appendix C), but must always comply with Layer 0.

1. Multi-Clock State Vector Model

[L1] SGTx maintains a multi-clock state vector for every trade. Each domain has its own logical clock that advances independently, enabling the platform to reason about temporal consistency across heterogeneous processes that operate at different speeds and with different finality guarantees.

The state vector consists of 12 domain clocks, each tracking the progress of a specific aspect of the trade lifecycle. The divergence index measures how far apart the clocks have drifted, which is a leading indicator of trade health and a trigger for Governor intervention.

Domain	Clock Tracks	Finality Classes
1. Commercial	Trade initiation, quote, contract, order	F0–F5
2. Logistics	Transport booking, execution, tracking, delivery	F0–F5
3. Customs	Declaration, assessment, clearance, post-clearance	F0–F5
4. Financial	Payment instructions, settlement, reconciliation	F0–F5
5. Documentation	Document generation, verification, legalization	F0–F5
6. Compliance	Sanctions, regulatory, KYB/AML screening	F0–F5
7. Insurance	Policy issuance, claims, settlement	F0–F5
8. Quality Control	Inspection, sampling, reporting	F0–F5
9. Dispute	Filing, mediation, arbitration, resolution	F0–F5
10. Post-Trade	Returns, claims, warranty, drawback	F0–F5
11. Evidence	Evidence package assembly, sealing	F0–F5
12. Governance	Governor decisions, OPA evaluations, Loom entries	F0–F5

Finality Classes (F0–F5): F0=None (no activity), F1=Proposed (intent expressed), F2=Asserted (action taken, awaiting confirmation), F3=Confirmed (counterparty or authority confirmed), F4=Settled (financial settlement complete), F5=Sealed (immutable, evidence-sealed, post-closure observation active).

Divergence Index: NONE (all clocks within 1 finality step), LOW (max 2 steps), MEDIUM (max 3 steps), HIGH (max 4 steps), CRITICAL (5+ steps — Governor intervention required).

2. Event Spine (Immutable Event Log)

[L1] The Event Spine is the immutable, append-only, hash-chained log of every significant event in the platform. It is the single source of truth for what happened, when, and by whose authority. The Event Spine is never modified or deleted; corrections are made by appending new events that reference the original.

Command ≠ Event: A Command is an intent to change state (e.g., 'SubmitQuote'). An Event is a fact that has occurred (e.g., 'QuoteSubmitted'). Commands flow into the Governor pipeline; Events are appended to the Event Spine after the Governor authorises the Command.

Observation / Assertion / Confirmation taxonomy: Events are classified by their authority attribute: Observation (passive recording of external state), Assertion (active claim by a party), Confirmation (authoritative verification by a trusted source). Each event type has a required authority class that determines who may emit it.

SHA-256 hash chain: Each event's hash is computed over (previousEventHash + eventPayload + timestamp + actorGtid). This creates a tamper-evident chain where any modification to a historical event breaks all subsequent hashes.

Replay mode (§86): The state vector for any USTN can be reconstructed by replaying all events from the Event Spine. This enables audit, dispute resolution, and regulatory inspection without querying live state.

3. Governor Pipeline

[L1] The Governor is the constitutional enforcement engine. Every irreversible action passes through the Governor pipeline, which evaluates the action against G1–G7 principles, OPA policies, and WasmEdge constitutional rules.

Pipeline stages: (1) Command received. (2) Governor evaluates G1 (execution gated). (3) OPA evaluates authored policies. (4) WasmEdge evaluates constitutional rules. (5) Multisig check if irreversible (G5). (6) Decision merged (DENY > CONDITIONAL > ALLOW). (7) If ALLOW, action executed + event appended to Loom. (8) If DENY, blocked + reason logged. (9) If CONDITIONAL, conditions recorded + action proceeds with conditions.

Decision merge semantics: When multiple gates evaluate the same action, the strictest verdict wins. DENY overrides CONDITIONAL overrides ALLOW. This ensures that a single gate's denial cannot be overridden by other gates' approvals.

4. Settlement Orchestration Control Plane

[L1] The Settlement Orchestration Control Plane manages multi-leg payment instructions. It orchestrates the flow of payment instructions to banks and PSPs, tracks settlement leg states, and enforces atomicity policies.

Multi-leg model: A single trade may have multiple payment legs (e.g., goods payment, freight payment, duty payment, SGTX fee). Each leg is an independent payment instruction with its own state machine: PENDING → SUBMITTED → ACCEPTED → SETTLED (or REJECTED / RETURNED / REVERSED).

5 atomicity policies: ALL_OR_NONE (all legs must settle or all fail), PARTIAL_ALLOWED (legs may settle independently), SEQUENCED (legs must settle in order), CONDITIONAL (legs settle based on conditions), HUMAN_RELEASE (legs require human release before settlement).

Bank Settlement Gateway (6-stage pipeline): (1) Schema validation (ISO 20022 / bank-specific format). (2) Signature validation (QES or bank API auth). (3) USTN validation (trade exists, not blocked). (4) Beneficiary consistency (match trade contract). (5) Bank policy check (limits, AML, sanctions). (6) AML/sanctions screening. The Gateway simulates this pipeline; real bank APIs are called by the connected bank's integration.

Non-custody boundary: SGTX orchestrates settlement instructions but NEVER holds funds. The FeeLock mechanism locks the SGTX fee at trade initiation, but the fee is collected by the bank during settlement — SGTX never receives the fee directly. This is the architectural non-custody boundary.

5. 28-Add-On Catalogue with Status Matrix

[L1] The SGTX platform includes 28 add-on modules. Each add-on extends the core platform with specific capabilities. The status matrix below uses the v14.0 deployment-state vocabulary: CORE_READY (implemented and tested), PRODUCTION_CONNECTED (live integration active), LEGAL_AUTHORIZATION_REQUIRED (technical ready, awaiting legal/regulatory approval).

#	Add-On Name	Priority	Status	Blueprint Ref
1	GNN Risk Engine & Institutional Trade Graph	Foundation	CORE_READY	Part 11.1
2	Federated Learning Network	Foundation	CORE_READY	Part 11.2
3	Causal Inference Engine	Foundation	CORE_READY	Part 11.3
4	Self-Healing Infrastructure & Chaos Engineering	Foundation	CORE_READY	Part 11.4
5	Automated Penetration Testing	Foundation	CORE_READY	Part 11.5
6	Post-Quantum Cryptography (PQC)	Foundation	CORE_READY	Part 11.6
7	Expanded ZK Proofs & Proof of Reserves	Foundation	CORE_READY	Part 11.7
8	Customs Bond & Guarantee Management	P0	CORE_READY	Part 11.8
9	Demurrage & Detention Management	P0	CORE_READY	Part 11.9
10	Broker Liability & Insurance Management	P0	CORE_READY	Part 11.10
11	Customs Valuation Intelligence	P1	CORE_READY	Part 11.11
12	Cold Chain Quality Management	P1	CORE_READY	Part 11.12
13	Inspection Agency Accreditation	P1	CORE_READY	Part 11.13
14	Currency Risk Management	P1	CORE_READY	Part 11.14
15	Government API Sandbox	P1	CORE_READY	Part 11.15
16	FTA Preference Management	P1	CORE_READY	Part 11.16
17	Piracy & Security Risk Engine	P1	CORE_READY	Part 11.17
18	Trade Compliance Calendar	P1	CORE_READY	Part 11.18
19	Cargo Insurance Integration	P2	CORE_READY	Part 11.19
20	Trade Finance Documentation	P2	CORE_READY	Part 11.20
21	Back-to-Back LC Management	P2	CORE_READY	Part 11.21
22	Force Majeure Handling	P2	CORE_READY	Part 11.22
23	Shipper's Declaration & Export Docs	P2	CORE_READY	Part 11.23
24	Port & Terminal Integration	P2	CORE_READY	Part 11.24
25	Payment Guarantee Confirmation (Optional)	P3	CORE_READY	Part 11.25
26	Demurrage Dispute Resolution	P3	CORE_READY	Part 11.26
27	(RESERVED)	—	—	—
28	GRiRE (Global Regulatory Intelligence & Requirements Engine)	Foundation	CORE_READY	Part 11.28

Non-marketplace bound: Add-On 1 (GNN) is explicitly bounded by the non-marketplace safeguard [L0-25]. It provides trust analytics for parties already known to the tenant but NEVER discovers, recommends, ranks, or introduces counterparties.

6. Jurisdiction Capability Adapter Schema

[L1] SGTX supports a worldwide jurisdiction capability adapter fabric. Each jurisdiction (country, customs territory, economic union) has a profile that defines its regulatory requirements, customs procedures, payment rails, and government integrations.

16 connector states: NOT_DISCOVERED → DISCOVERED → DOCUMENTED → CONTACT_REQUIRED → CREDENTIALS_REQUIRED → SANDBOX_AVAILABLE → SANDBOX_CONNECTED → CERTIFICATION_REQUIRED → CERTIFICATION_PENDING → PRODUCTION_READY → PRODUCTION_CONNECTED → DEGRADED → OUTAGE → PORTAL_ONLY → MANUAL_ONLY → DEPRECATED.

7 authoritative statuses: SGTX_READY → SUBMITTED → GOVERNMENT_ACCEPTED → GOVERNMENT_REJECTED → GOVERNMENT_HOLD → GOVERNMENT_RELEASED → MANUAL_AUTHORITY_CONFIRMED. SGTX never invents government approval.

4-dimension external readiness: Every integration reports independently across TECHNICAL (API connected?), LEGAL (contracts signed?), OPERATIONAL (procedures tested?), COMMERCIAL (fees agreed?). 'Connected' requires all four dimensions to be satisfied.

7. Regulatory Classification Gate

[L1] The Regulatory Classification Gate maps the platform's actual functionality to the applicable licence class in each jurisdiction. This is a dynamic, jurisdiction-aware classification — not a static label.

[L1] **Non-custody is architectural, not legal classification.** SGTX's non-custody property is an architectural fact (the platform never holds customer funds). However, actual functionality determines licensing in each jurisdiction. A jurisdiction may classify SGTX's orchestration role as requiring specific licences (e.g., payment service provider, money transmission, customs broker licence) regardless of the non-custody architecture.

Egypt / CBE Law 194/2020 as initial reference: The platform's initial regulatory classification is based on Egyptian law, including CBE Law 194/2020, Nafeza customs regulations, and ETA (Egyptian Tax Authority) requirements. The classification matrix is extensible to other jurisdictions via the Jurisdiction Capability Adapter.

8. Closure Policy

[L1] A USTN is closed only when all 7 closure conditions are met. Closure is earned, not forced.

#	Condition	Description
1	Delivery Accepted	Final delivery confirmed by receiver (quantity, condition, quality verified)
2	Settlement Complete	All payment legs settled (goods, freight, duty, fees)
3	Financial Reconciliation Complete	Bank/PSP reconciliation matches SGTX records (no material mismatch)
4	Customs Complete	Import and export customs cleared, no open holds
5	Post-Clearance Complete	Post-clearance audit, correction, refund/drawback processed if applicable
6	Disputes/Claims Satisfied	All filed disputes and claims resolved or time-barred
7	Evidence Sealed	26-category evidence package assembled, hashed, and sealed (immutable)

canClose predicate: `canClose(ustn) = AND(all 7 conditions met)`. The Governor evaluates this predicate; it never auto-closes. If `canClose` returns true, the USTN is marked CLOSED and the evidence package is sealed.

CLOSED_WITH_EXCEPTION: In limited cases, a USTN may be closed with an open exception if the sole blocker is an EXCEPTION_OPEN state with severity ≤ 2 (low). Severity 3–5 exceptions block closure. The severity matrix ensures that minor administrative exceptions do not indefinitely block closure, while material exceptions do.

Post-closure event rules (§22): After closure, a post-closure observation period is active. New evidence may be added (e.g., late-arriving customs confirmation) but sealed evidence cannot be modified. The Transaction Twin continues to observe and may trigger re-opening if a material exception is discovered.

9. AI Recommendation Gateway

[L1] The AI Recommendation Gateway channels AI outputs through the Governor pipeline. AI never executes directly; it proposes, and the Governor decides.

Flow: (1) AI subsystem (A1–A3) produces a recommendation. (2) Recommendation is submitted to the Governor as a Command. (3) Governor evaluates against G1–G7, OPA policies, WasmEdge rules. (4) If ALLOW, the recommended action is executed by the deterministic policy engine (A4). (5) If CONDITIONAL, conditions are recorded. (6) If DENY, the recommendation is blocked + reason logged. (7) The event is appended to the Loom.

[L0-9, L0-21] **AI NEVER has independent execution authority.** A4 is deterministic policy automation — the AI proposes, the Governor + OPA + WasmEdge execute under pre-authorised rules. The AI subsystem itself never acquires execution authority. This is the constitutional boundary.

10. Transport Engine Architecture

[L1] SGTX supports 5 transport modes as first-class engines, each with its own entity types, state machines, and terminal adapters. A Multimodal Orchestrator coordinates multi-leg journeys across modes.

Mode	Entity Types	State Machine	Portal
Road	RoadCorridor, RoadLeg, RoadShipment, RoadVehicle, RoadDriver, RoadBorderCrossing, RoadGpsTracking	PLANNED → IN_TRANSIT → AT_BORDER → CLEARED → DELIVERED	LSP
Air	AirBooking, AirFlight, AirAirport, AirWaybill, AirPiece, AirUld, AirStatusEvent, AirChargeableWeight	BOOKED → DOCUMENTS_PENDING → CUSTOMS_PENDING → READY_FOR_GATE → GATE_IN → INSPECTED → YARD → LOADED → AT_SEA → DISCHARGED → DELIVERED	Shipping Line
Ocean Container	Booking, Vessel, Voyage, Port, Container, VGM, B/L, e-B/L, Manifest, ACI, Terminal, Gate, Transshipment, Demurrage, Detention, Delivery	PLANNED → LOADED → DEPARTED → IN_TRANSIT → ARRIVED → DISCHARGED → CUSTOMS_RELEASED → DELIVERED	Shipping Line
RoRo	RoRoShipment, RoRoUnit (VIN-level), RoRoVoyage, RoRoBooking, RoRoYard, RoRoGateEvent, RoRoInspection, RoRoBillOfLading	19-state unit machine: BOOKED → ... → DELIVERED → ACCEPTED	Shipping Line
Rail	RailBooking, RailTrain, RailWagon, RailTerminal, RailConsignment (CIM/SMGS), RailTransit, RailStatusEvent	BOOKED → LOADED → DEPARTED → AT_BORDER → CUSTOMS_RELEASED → ARRIVED → UNLOADED → DELIVERED	LSP

Multimodal Orchestrator: A single USTN may span multiple transport modes (e.g., Truck → RoRo → Rail → Truck). The Multimodal Orchestrator coordinates the handoff between modes, ensuring that the USTN remains the canonical reference across all legs.

LAYER 2 — IMPLEMENTATION SPECIFICATIONS

[L2] This layer contains concrete, testable implementation artefacts. These specifications are designed to be directly implementable by an engineering team without ambiguity. They include canonical data models, event type catalogues, API contracts, and observability requirements.

11. Canonical Event Type Catalogue

[L2] The following event types are canonical. Each event has required fields, an authority class (who may emit it), a clock impact (which domain clock it advances), and lineage rules (what it references).

Event Type	Authority Class	Clock Impact	Lineage
TradeInitiated	Assertion (Buyer)	Commercial F1	References: buyerGtid, sellerGtid
QuoteSubmitted	Assertion (Seller)	Commercial F2	References: ustn, tradeld
QuoteAccepted	Assertion (Buyer)	Commercial F3	References: ustn, quoteld
ContractLocked	Confirmation (Governor)	Commercial F4, Governance F3	References: ustn, multisigId
RegulatorySnapshotCaptured	Observation (System)	Compliance F3	References: ustn, hash
ShipmentBooked	Assertion (Shipper)	Logistics F2	References: ustn, bookingRef
ContainerLoaded	Observation (Terminal)	Logistics F3	References: ustn, containerNumber
VesselDeparted	Observation (Carrier)	Logistics F3	References: ustn, voyageNumber
CustomsDeclarationSubmitted	Assertion (Broker)	Customs F2	References: ustn, declarationRef
CustomsCleared	Confirmation (Authority)	Customs F4	References: ustn, clearanceRef
PaymentInstructionSubmitted	Assertion (Payer)	Financial F2	References: ustn, legId
PaymentSettled	Confirmation (Bank)	Financial F4	References: ustn, legId, bankRef
InspectionCompleted	Assertion (QC/Lab)	QC F3	References: ustn, inspectionId
DeliveryAccepted	Assertion (Receiver)	Logistics F5	References: ustn, podRef
DisputeFiled	Assertion (Party)	Dispute F2	References: ustn, disputeld
USTNClosed	Confirmation (Governor)	All domains F5	References: ustn, evidenceHash
ExceptionRaised	Observation (System)	Governance F2	References: ustn, exceptionId
RecoveryExecuted	Confirmation (Governor)	Governance F3	References: ustn, exceptionId, recoveryPath

Required fields for ALL events: eventId, eventType, ustn, timestamp, actorGtid, authorityClass, previousEventHash, payload (JSON), signature (QES or system). Events without these fields are rejected.

12. Canonical Data Model Outline

[L2] The SGTX Canonical Data Model consists of the following primary stores:

Store	Purpose	Key Fields
State Vector Store	12-domain × F0-F5 finality tracking per USTN	ustn, domainClocks[12], finalityClass, divergenceIndex, healthScore
Event Spine	Immutable hash-chained event log	eventId, previousEventHash, eventType, ustn, timestamp, actorGtid, payload, signature
Obligation Graph	Directed dependency graph of trade obligations	obligationId, ustn, type, state, dependencies[], transitiveImpact
Settlement Instructions/Legs	Multi-leg payment instruction tracking	instructionId, ustn, legs[], atomicityPolicy, state
External Identifier Registry	17 identifier types (UCR, MAWB, HAWB, B/L, etc.)	identifierId, type, value, ustn, lifecycle, issuingAuthority
Recovery Vault	Content-addressable (SHA-256) evidence storage	entryId, ustn, type, contentHash, reference, verified
Transaction Twin	14-domain digital twin for post-closure observation	twinId, ustn, domains[14], postClosureActive, observationExpiry
Financial Exposure	14-dimension exposure tracking per USTN	exposureId, ustn, dimensions[14], state, outstandingAmount
Exception Events	Severity 1-5 exception tracking with SLA	exceptionId, ustn, category, severity, slaDeadline, resolutionAction
Closure Policy	7-condition closure evaluation per USTN	policyId, ustn, conditions[7], blockers[], canClose, closureState

13. API Contract Structure

[L2] The SGTX API enforces Command/Event separation. Commands are POST/PUT requests that intent to change state. Events are GET requests that retrieve immutable records. No API endpoint both reads and writes state.

Command endpoints (POST/PUT): /api/sgtx/trade-request (create trade), /api/sgtx/quote/submit (submit quote), /api/sgtx/contract/lock (lock contract), /api/sgtx/ustn-close (close USTN), etc. All commands are idempotent (via Idempotency-Key header) and rate-limited.

Event/query endpoints (GET): /api/sgtx/trade/[ustn] (fetch trade), /api/sgtx/events/[ustn] (event history), /api/sgtx/regulatory-snapshot/[ustn] (fetch snapshot), etc. All queries are read-only and cached where appropriate.

Standard response envelope: All API responses use { ok: boolean, data: ..., error?: string, filter?: object }. List endpoints return { ok, , count, filter }. This is the v14.0 standardized contract.

14. Observability Catalogue

[L2] SGTX maintains the following observability surfaces:

Surface	Metrics	Retention
Platform Health	Request rate, error rate, latency p50/p95/p99, uptime	30 days
Trade Metrics	Trades initiated, USTN closure rate, avg trade value, avg time-to-close	7 years
Governor Metrics	Decisions per minute, DENY rate, CONDITIONAL rate, multisig latency	7 years
Event Spine Metrics	Events per minute, hash chain verification status, replay success rate	7 years
Settlement Metrics	Settlement leg states, bank API latency, reconciliation match rate	7 years
Exception Metrics	Open exceptions by severity, SLA breach count, recovery success rate	7 years
AI Metrics	Recommendations per minute, acceptance rate, fallback rate, provider latency	30 days
Integration Metrics	Connector health (16 states), API success rate, sandbox vs production	90 days

Minimum failure-path test suite: The platform maintains an adversarial test suite covering: (1) USTN closure with missing conditions (must block). (2) Non-custody violation attempt (must block). (3) AI A5 autonomy attempt (must block). (4) Marketplace matching attempt (must block). (5) Event spine tampering (must detect). (6) Governor bypass attempt (must block). (7) Settlement without bank confirmation (must block). (8) Recovery that erases history (must block).

15. RTO/RPO Targets

[L2] Recovery Time Objective and Recovery Point Objective targets:

Tier	RTO	RPO	Scope	Durability
Critical	4 hours	0 (append-only)	Event Spine, State Vector, Governor decisions	3 copies, 2 jurisdictions, quorum 2-of-3
Standard	24 hours	0 (append-only)	All trade data, settlement instructions, evidence	3 copies, 2 jurisdictions, quorum 2-of-3
Extended	72 hours	< 1 hour	Add-on data, analytics, logs	2 copies, 1 jurisdiction, single-write

Durability definition: '3 copies, 2 jurisdictions, quorum 2-of-3' means data is replicated to 3 storage nodes across at least 2 legal jurisdictions, and writes require acknowledgement from 2 of 3 nodes before acknowledgement. This ensures no single jurisdiction's legal action can make data unavailable.

APPENDICES

Appendix A — Financial Control Framework (CFO)

[L2] This appendix defines the financial control framework for SGTX, including fee schedule, reserve policy, non-custody attestation, and unit economics.

A.1 Fee Schedule

SGTX Platform Fee: 1.5% of trade value, collected at settlement via the bank (non-custodial). The fee is locked at trade initiation (FeeLock) and collected by the bank during settlement — SGTX never receives the fee directly.

Optional service fees: QC inspection (buyer-requested), lab tests (buyer-requested), customs broker services, logistics RFQ fees. These are quoted by the service provider and accepted by the trader; SGTX does not set these prices (non-marketplace principle).

A.2 Reserve Policy

[L0-17] **110% reserve rule:** If reserve metadata is maintained, it must be at least 110% backed. The constitutional layer sets the threshold; ZK attestation (Add-On 7) provides cryptographic evidence that the threshold is met.

Composition: Reserves may consist of cash, bank guarantees, government bonds, or other approved instruments. The composition is jurisdiction-aware (each jurisdiction may have its own approved instrument list).

ZK attestation: Add-On 7 generates zero-knowledge proofs that the reserve ratio is $\geq 110\%$ without revealing the exact amount or composition. This protects institutional privacy while proving compliance.

A.3 Non-Custody Attestation Template

SGTX provides a standard non-custody attestation template for use in regulatory filings and bank onboarding: 'SGTX Platform does not hold, receive, or transfer customer funds at any point in the trade lifecycle. All funds flow through connected banks and regulated payment service providers. SGTX orchestrates payment instructions but never becomes the settlement authority. The FeeLock mechanism locks the platform fee at trade initiation, but the fee is collected by the bank during settlement.'

Appendix B — Regulatory & Legal

B.1 Regulatory Classification Matrix

[L2] The Regulatory Classification Matrix maps SGTX functionality to licence classes per jurisdiction. The initial matrix covers Egypt; it is extensible to other jurisdictions via the Jurisdiction Capability Adapter.

Functionality	Egypt Classification	Required Licence	Status
Trade orchestration	Service provider	None (non-regulated)	OPERATIONAL
Payment instruction	Payment service provider	CBE registration (if applicable)	LEGAL_AUTHORIZATION_REQUIRED
Customs declaration	Customs broker facilitation	Broker licence (broker-side)	OPERATIONAL
Document issuance	Digital document service	ETA/CargoX integration	CORE_READY
Settlement orchestration	Non-custodial orchestration	Bank partnership	LEGAL_AUTHORIZATION_REQUIRED
AI advisory	AI service provider	None (advisory only)	OPERATIONAL

B.2 Data-Residency Element Classification

[L2] Data elements are classified by residency requirement: EGYPT_ONLY (must stay in Egypt), COUNTRY_ONLY (must stay in origin/destination country), REGIONAL (may flow within a region), APPROVED_CROSS_BORDER (may flow to approved jurisdictions), GLOBAL_ALLOWED (no restriction). The strictest applicable rule wins.

B.3 SAR/FIU Interface Notes

[L2] SGTX maintains interface notes for Suspicious Activity Report (SAR) / Financial Intelligence Unit (FIU) reporting. The platform does not file SARs directly (that is the bank's responsibility), but it provides structured evidence packages that banks can use for SAR filing. The Add-On 17 (Piracy & Security Risk Engine) and Add-On 7 (ZK Proofs) support the evidence chain.

Appendix C — Operating Model

C.1 Platform Governance Authority RACI

Decision	Responsible	Accountable	Consulted	Informed
Constitutional amendment	Platform Governance Authority	Constitutional Signatories (3-of-5)	Legal, Compliance, Engineering	All stakeholders
Policy update (OPA)	Policy Team	Governance Authority	Engineering, Compliance	Operations
WASM module update	Engineering	CTO	Security, Governance	Operations
Integration activation	Integration Team	CTO	Legal, Compliance	Operations
Incident response (SEV-0)	On-call Engineer	CTO	Security, Legal, Comms	All stakeholders
Incident response (SEV-1)	On-call Engineer	Engineering Lead	Security	Operations

C.2 Multisig Policy

[L2] **Standard multisig:** 2-of-3 for irreversible trade actions (USTN closure, contract lock, settlement release). **Constitutional multisig:** 3-of-5 for platform-level changes (policy amendment, WASM module update, integration activation). Signatories are identified by GTID and must use Qualified Electronic Signatures (QES).

C.3 Change-Management Process

[L2] WASM module and policy changes follow a 4-stage process: (1) Author (draft policy/module). (2) Review (security audit + legal review). (3) Approve (multisig 3-of-5). (4) Deploy (staged rollout: sandbox → production). Each stage is Loom-logged. Rollback is always possible for policies (previous version is retained); WASM modules are versioned with backward compatibility.

C.4 Incident Severity Model

Severity	Definition	Response Time	Escalation
SEV-0	Platform-wide outage or data breach	Immediate (24/7)	CTO + Legal + Comms + Board
SEV-1	Major feature failure affecting multiple trades	1 hour	Engineering Lead + CTO
SEV-2	Feature failure affecting single trade or tenant	4 hours	On-call Engineer
SEV-3	Minor bug with workaround	24 hours	Assigned Engineer
SEV-4	Cosmetic or enhancement request	Next sprint	Product Backlog

Appendix D — Implementation Priority Framework

[L2] Implementation is organised into P0–P4 waves. These are dependency-driven, not calendar-driven. No contractual calendar claims are made — each wave is completed when its dependencies are satisfied.

Wave	Scope	Dependencies	Status
P0 — Core Trade Execution	Trade, Contract, USTN, Governor, Event Spine, State Vector, Closure Policy	None (foundation)	CORE_READY
P1 — Transport Engines	Road, Air, Ocean Container, RoRo, Rail, Multimodal Orchestrator	P0 complete	CORE_READY
P2 — Intelligence & Compliance (Add-Ons 1-8)	GNN, Federated Learning, Causal Inference, Self-Healing, Pentest, PQC, ZK, Customs Bond	P0 complete	CORE_READY
P3 — Operational Add-Ons (Add-Ons 9-18)	Demurrage, Broker Liability, Valuation, Cold Chain, Inspection, Currency Risk, Gov Sandbox, FTA, Security, Compliance Calendar	P0 + GRiRE (P4 foundation)	CORE_READY
P4 — Extended Functionality (Add-Ons 19-28)	Cargo Insurance, Trade Finance, Back-to-Back LC, Force Majeure, Export Docs, Port/Terminal, Payment Guarantee, Demurrage Dispute, GRiRE	P0 + P2 (for intelligence dependencies)	CORE_READY

Appendix E — Full Historical Audit Trail (Immutable Annex)

This annex preserves the full audit trail from v13.1 Part A (findings A-01 through A-24) and the v12.0 audit header. Nothing has been removed. The audit findings are the governing resolutions for all v14.0 architecture.

E.1 v13.1 Part A Audit Summary (A-01 through A-24)

Findin g	Conflict	Resolution (Governing)
A-01	Direct Government API Scope	Four Egypt connectors (Nafeza, CargoX, ETA, CBE) are the initial reference set; worldwide adapter fabric is the extensibility layer.
A-02	AI A4 Authority Language	A4 is deterministic policy automation by Governor+OPA+WasmEdge; AI never has independent execution authority.
A-03	Non-Custody vs Payment	Non-custody is architectural; Bank Settlement Gateway orchestrates but never holds funds.
A-04	Stablecoin/DeFi vs Bank Settlement	DeFi is conditional, jurisdiction-permitted sub-rail; bank-authoritative settlement is canonical.
A-05	Reserve Metadata vs Custody	Reserve tables store attestations only; they do not create custody.
A-06	Reserve Ratio Thresholds	110% rule is constitutional; ZK attestation provides evidence. Both preserved.
A-07	GNN Non-Marketplace	Trust graph is relationship analytics for known parties; never discovers/recommends/ranks.
A-08	Seven vs Twenty-Eight Add-Ons	Original seven are historical reference; full 28-add-on catalogue is canonical.
A-09	RoRo as First-Class	RoRo is a first-class transport mode, not an ocean sub-mode.
A-10	Egypt Nafeza Mode-Specific	Mode-specific applicability architecture; not one generic maritime workflow.
A-11	Closure Semantics	7-condition closure; CLOSED_WITH_EXCEPTION for severity ≤ 2 only.
A-12	USTN as Namespace	USTN is universal reference; does not override external authorities.
A-13	ISO 20022 as Output	ISO 20022 is an integration output format, not canonical data.
A-14	Egypt Data Localisation	EGYPT_ONLY for specific elements; strictest applicable rule wins.
A-15	Zero-Cost Clarified	Institutional-cost scope: data sources are free; institutional costs (bank fees, broker fees, integration costs) are real.
A-16	Marketplace Terminology	SGTX is relationship-controlled orchestration, NOT a public marketplace.
A-17	AI Explanations vs Decisions	AI generates explanations; Governor makes authoritative decisions.
A-18	Manual Fallback	Manual fallback is governed: authenticated, attributable, timestamped, Loom-logged.
A-19	Global Readiness Claims	Use CORE_READY/PRODUCTION_CONNECTED vocabulary; never WORLDWIDE_INTEGRATED.
A-20	Command vs Event	Command (intent) $\neq$ Event (fact). Commands flow to Governor; Events append to Spine.
A-21	Recovery vs Rollback	Recovery restores state; never erases history. Loom is immutable.
A-22	Proof-of-Reserves Wallet Examples	Wallet/bank examples are verification-only; consistent with non-custody.
A-23	Jurisdiction Examples	Country-specific examples are illustrative configuration governed by active jurisdiction profile.
A-24	Incomplete Change-Set	Add-Ons 9-28 and Final Summary now fully integrated into v14.0.

Appendix F — Source Manifest

This document was produced from the following sources. SHA-256 hashes are computed over the extracted text content.

Source	Type	Lines	SHA-256
SGTX_PLATFORM_MASTER_BLUEPRINT_INTEGRATED_v12.0.docx	Main Blueprint	~76,122	c263f527f3966dab01d3cffc87e7d2747d01c017ca7a786d1964f09018087d42
sgtx add ons and modifications.rtf	Change-Set	~7,582	87181d220df82a485485eec4b9896031910c79afa6332516ef2afac4682c5b72
SGTX_v13.1_FINAL.docx	Integrated Edition	~47,216	(computed from extracted text)

Change Log v13.1 → v14.0

Change	Rationale
Three-layer separation (L0/L1/L2)	Separate immutable principles from mutable architecture from testable specs
Over-claim elimination	Replaced 'production-ready' with CORE_READY; 'complete' with 'specified'; 'zero-cost' with 'institutional-cost scope clarified'
28-add-on status matrix	Explicit deployment-state vocabulary for every add-on
4-dimension external readiness	TECHNICAL/LEGAL/OPERATIONAL/COMMERCIAL reported independently per connector
Command/Event taxonomy formalized	Command (intent) ≠ Event (fact) — enforced in API contract
32-point constitution	All 32 constitutional points listed explicitly with [L0] tags
Audit trail preserved as annex	A-01 through A-24 preserved verbatim as governing resolutions
Dependency graph added	Visual cross-reference of major components
RTO/RPO with quorum definitions	Explicit durability/quorum requirements per tier
No calendar claims	P0–P4 waves are dependency-driven, not date-driven